

3453. Separate Squares I

Solved 

Medium

 Topics

 Companies

 Hint

You are given a 2D integer array `squares`. Each `squares[i]` = `[xi, yi, li]` represents the coordinates of the bottom-left point and the side length of a square parallel to the x-axis.

Find the **minimum** y-coordinate value of a horizontal line such that the total area of the squares above the line *equals* the total area of the squares below the line.

Answers within `10-5` of the actual answer will be accepted.

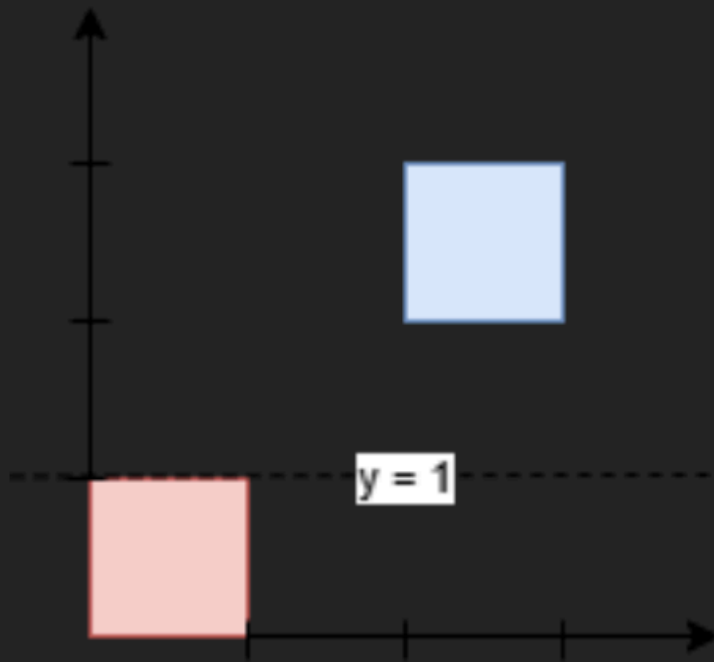
Note: Squares **may** overlap. Overlapping areas should be counted **multiple times**.

Example 1:

Input: `squares = [[0,0,1],[2,2,1]]`

Output: 1.00000

Explanation:



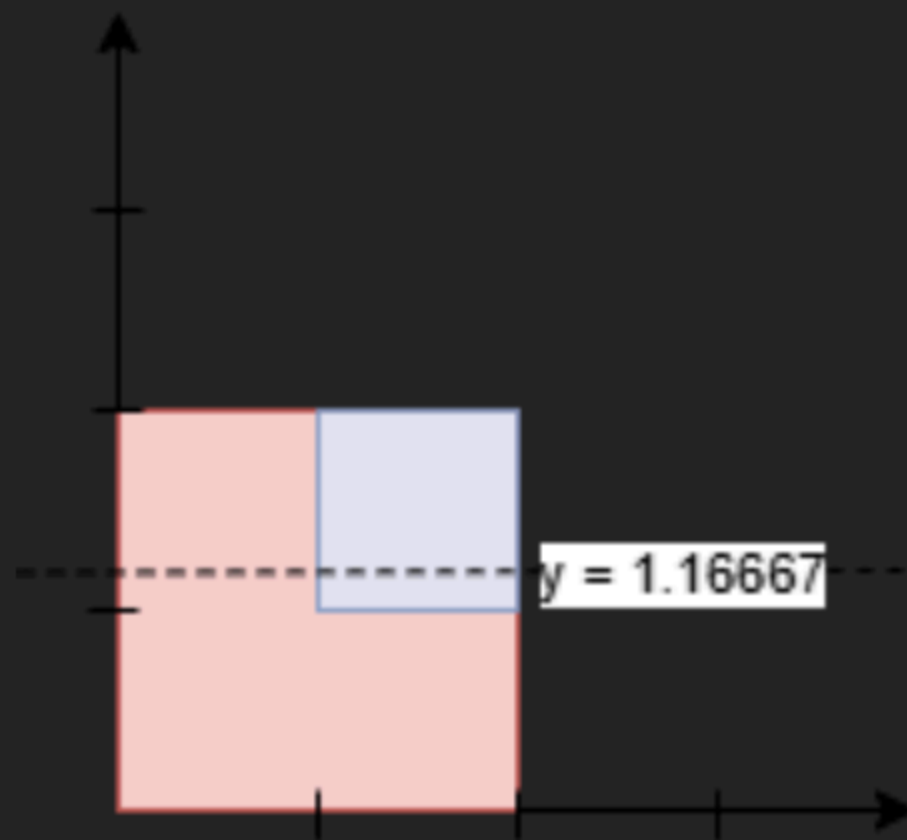
Any horizontal line between `y = 1` and `y = 2` will have 1 square unit above it and 1 square unit below it. The lowest option is 1.

Example 2:

Input: squares = `[[0,0,2],[1,1,1]]`

Output: 1.16667

Explanation:



The areas are:

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- Below the line: $\frac{7}{6} * 2$ (Red) + $\frac{1}{6}$ (Blue) = $\frac{15}{6} = 2.5$.
- Above the line: $\frac{5}{6} * 2$ (Red) + $\frac{5}{6}$ (Blue) = $\frac{15}{6} = 2.5$.

Since the areas above and below the line are equal, the output is $\frac{7}{6} = 1.16667$.

Constraints:

- $1 \leq \text{squares.length} \leq 5 * 10^4$
- $\text{squares}[i] = [x_i, y_i, l_i]$
- $\text{squares}[i].\text{length} == 3$
- $0 \leq x_i, y_i \leq 10^9$
- $1 \leq l_i \leq 10^9$
- The total area of all the squares will not exceed 10^{12} .

Python:

class Solution:

def separateSquares(self, squares: List[List[int]]) -> float:

axis = Counter()

total = 0

for _, y, s in squares:

area = s*s

total += area

axis[y] += s

axis[y+s] -= s

target = total / 2

events = sorted(axis.items())

cur = 0

last_y = events[0][0] - 1

for y, a_unit in events:

dy = (y-last_y)

nar = cur * dy

```
if target - nar <= 0:
    return last_y + (target / nar) * dy
```

```
target -= nar
cur += a_unit
last_y = y
```

JavaScript:

```
function helper(line, squares) {
    let aAbove = 0, aBelow = 0;
    let n = squares.length;

    for (let i = 0; i < n; i++) {
        let x = squares[i][0], y = squares[i][1];
        let l = squares[i][2];
        let total = l * l;

        if (line <= y) {
            aAbove += total;
        } else if (line >= y + l) {
            aBelow += total;
        } else {
            let aboveHeight = (y + l) - line;
            let belowHeight = line - y;
            aAbove += l * aboveHeight;
            aBelow += l * belowHeight;
        }
    }

    return aAbove - aBelow;
}
```

```
var separateSquares = function(squares) {
    let lo = 0, hi = 2e9;

    for (let i = 0; i < 60; i++) {
        let mid = (lo + hi) / 2.0;
        let diff = helper(mid, squares);

        if (diff > 0) lo = mid;
        else hi = mid;
    }

    return hi;
}
```

```
};
```

Java:

```
class Solution {
    private double helper(double line, int[][] squares) {
        double aAbove = 0, aBelow = 0;
        for (int i = 0; i < squares.length; i++) {
            int x = squares[i][0], y = squares[i][1], l = squares[i][2];
            double total = (double) l * l;

            if (line <= y) {
                aAbove += total;
            } else if (line >= y + l) {
                aBelow += total;
            } else {
                // The line intersects the square.
                double aboveHeight = (y + l) - line;
                double belowHeight = line - y;
                aAbove += l * aboveHeight;
                aBelow += l * belowHeight;
            }
        }
        return aAbove - aBelow;
    }

    public double separateSquares(int[][] squares) {
        double lo = 0, hi = 2*1e9;

        for (int i = 0; i < 60; i++) {
            double mid = (lo + hi) / 2.0;
            double diff = helper(mid, squares);

            if (diff > 0)
                lo = mid;
            else
                hi = mid;
        }

        return hi;
    }
}
```